

Summer Analytics 2026

Mini-Hackathon - Week 2

E-Commerce Conversion Prediction Challenge

Overview

Welcome to the Summer Analytics 2026 Hackathon.

This challenge focuses on predictive modeling using structured tabular data. Participants are expected to analyze the dataset, perform appropriate preprocessing, engineer meaningful features, and build machine learning models to predict customer conversion outcomes.

The competition is designed to assess:

- Exploratory Data Analysis (EDA)
- Data Preprocessing
- Feature Engineering
- Classification Modeling
- Model Evaluation

DATASET LINK : [Find Here](#)

Objective

Given the features provided in the dataset, predict whether a user converts.

Target Variable

Converted

- 1 = Converted
 - 0 = Not Converted
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Files Provided

train.csv

Contains feature columns along with the target variable.

public_test.csv

Contains feature columns along with the target variable.

Participants may use this dataset for validation and experimentation.

private_test.csv

Contains feature columns only.

Participants must generate predictions for this dataset.

sample_submission.csv

Contains the required submission format.

All relevant Files : [Click Here](#)

Dataset Information

The dataset consists of anonymized user-level observations.

Column	Description
User_ID	Unique record identifier
Age	Age of the user
Income	Estimated annual income

City_Tier	City classification
Device_Type	Device category
Traffic_Source	Acquisition source
Pages_Viewed	Number of pages viewed
Products_Viewed	Number of products viewed
Time_On_Site	Session duration
Previous_Purchases	Historical purchases
Discount_Seen	Discount visibility indicator
Browser_Version	Browser version identifier
Campaign_Code	Campaign identifier
Converted	Target variable (available only in train.csv and public_test.csv)

Important Note

The dataset has been designed to resemble real-world machine learning data.

Participants are encouraged to thoroughly explore the data and make informed decisions regarding preprocessing, feature engineering, model selection, and evaluation.

No additional information regarding dataset generation or hidden characteristics will be provided.

Evaluation Metric

Submissions will be evaluated using:

F1 Score

Final rankings will be determined using the F1 Score obtained on the hidden labels corresponding to `private_test.csv`.

Higher scores indicate better predictive performance.

Submission Format

Participants must submit a CSV file named:

`submission.csv`

The file must contain exactly two columns:

`User_ID,Converted`

Example:

`User_ID,Converted`

`103001,1`

`103002,0`

`103003,1`

Submission Requirements

- Column names must remain unchanged.
 - User_ID values must remain unchanged.
 - Predictions must be binary values (0 or 1).
 - Additional columns are not permitted.
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Deliverables

Participants must submit:

1. submission.csv

Predictions for all records in private_test.csv.

2. Jupyter Notebook (.ipynb) -> name it as notebook.ipynb

The notebook used to generate the final submission.

The notebook must be reproducible and correspond to the submitted predictions.

3. One-Page Report (.pdf)

A one-page summary of the methodology used to generate the final submission.

Failure to submit any required file may result in disqualification.

Starter Notebook

A starter notebook (starter_notebook.ipynb) has been provided.

The notebook demonstrates:

- Loading the datasets
- Basic preprocessing
- Training a simple baseline model
- Generating predictions
- Creating a valid submission file

Use of the starter notebook is optional.

Programming Language

Only Python may be used for model development and submission generation.

Use of AI Tools

The use of AI-assisted coding, machine learning, or problem-solving tools is strictly prohibited.

Participants are expected to independently perform all analysis, preprocessing, feature engineering, model development, debugging, and submission generation.

Use of AI-generated code, notebooks, reports, or solutions may result in disqualification.

Academic Integrity

Participants must not:

- Share code or predictions with other participants.
- Submit work that is not their own.
- Attempt to obtain hidden evaluation labels.
- Manipulate submission files.
- Use unauthorized external assistance.

Violations may result in immediate disqualification.

Verification Process

Organizers reserve the right to conduct verification checks on shortlisted participants.

Verification may include:

- Review of submitted notebooks
- Reproducibility checks
- Discussion of methodology and approach

Participants must be able to explain the methodology used in their submission.

Tie-Breaking Policy

In the event of identical F1 Scores, rankings will be determined using:

1. Earlier valid submission timestamp
 2. Successful verification of notebook and methodology
 3. Final decision of the organizing team
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Organizer Decisions

All decisions made by the organizing team regarding evaluation, rankings, eligibility, and rule enforcement shall be final and binding.

Good luck!

Summer Analytics 2026
Consulting & Analytics Club, IIT Guwahati